

Clifford Alg.:  $\{\gamma_m, \gamma_n\} = \gamma_m \gamma_n + \gamma_n \gamma_m = 2\eta_{mn} \mathbb{I}$   
 $m, n = 1, \dots, D$

$\eta_{mn}$  metric for  $\mathbb{R}^{s,t}$  ( $s+t=D$ )

- $\mathbb{I} \gamma_n = \gamma_n \mathbb{I} \quad \forall n = 1, \dots, D$   
 $\mathbb{I} = \mathbb{I}^2$

# These elements  $\{\gamma_1, \dots, \gamma_D, \mathbb{I}\}$  generate the group  $C_D$ :

$$C_D \equiv \{ \pm \mathbb{I}, \pm \gamma_m, \pm \gamma_{m_1 m_2}, \dots, \pm \gamma_{m_1 \dots m_D} \}$$

where  $\gamma_{m_1 \dots m_p} \equiv \gamma_{[m_1 \dots m_p]}$

$\Rightarrow$  Since  $\gamma_{m_1 \dots m_p}$  acts as on  $p$ -form we have

$$\binom{D}{p} = \frac{D!}{(D-p)! p!}$$

elements of rank  $p$ .

$\therefore$  We can easily determine the size of the set  $C_D$

$$\|C_D\| = 2 \sum_{p=0}^D \binom{D}{p}$$

$\uparrow$  factor of 2 from  $\pm$

Binomial expansion implies

$$(1+1)^D = \sum_{p=0}^D \binom{D}{p}$$

$$\|C_D\| = 2^{D+1}$$

Cliff. Alg. in even dim's:

Claim: the number of irreducible reps of any finite dim group  $G$  equals the number of its conjugacy classes.

Conjugacy Class:  $K_a$

$$K_a = \{ g \in G \mid H a g^{-1} = a \}$$

look at the orders of  $\gamma$ :

$$\underline{+1} \quad b = g(+1)g^{-1} = 1$$

$$\therefore 1 \sim I$$

$$\underline{-1} \quad b = g(-1)g^{-1} = -1$$

$$(-1) \sim (-I)$$

$\leadsto$  Each define a separate conjugacy class of order 1.

$$\underline{\gamma_m} \quad b = g \gamma_m g^{-1}$$

$$\gamma^m \gamma^n + \gamma^n \gamma^m = 2M^{mn} \gamma$$

$$2\gamma^n \gamma_n = 2\text{Tr}(M) \gamma$$

$$\gamma^n \gamma_n = D \gamma$$

$$\Rightarrow (\gamma^n)^{-1} = \left(\frac{1}{D}\right) \gamma_n$$

$\therefore$

$$b = \gamma^n \gamma_m \left[ \frac{1}{D} \gamma_n \right]$$

$$= \frac{1}{D} \gamma^n (-\gamma_n \gamma_m + 2M_{nm} \gamma)$$

$$= D^{-2} [\gamma^n (-\gamma_n) \gamma_m + 2\gamma_m]$$

$$= -\gamma_m + 2D \gamma_m$$

$$= (2D - 1) \gamma_m$$

$$\therefore \text{For } g \in \mathcal{O}(\gamma^2) \quad b \propto \gamma_m$$

All conjugacy classes of  $\mathcal{O}_D$  are therefore  $[+1], [-1], [\gamma_m], \dots, [\gamma_{m_1 \dots m_D}]$

where  $D \in 2\mathbb{Z}$

\* Note: Each class  $K_g$  is unique for each

$$[\gamma_{m_1 \dots m_i}]^{i.e.} [\gamma_{21}]$$

$$|P| = \frac{D!}{D_1!} = 1$$

$\therefore$

There are therefore

$$\sum_{p=0}^D (P) = 2^D$$

$$\text{for } [+1] \oplus [\gamma_m] \dots [\gamma_{m_1 \dots m_D}]$$

Including  $[-1]$  we have

$$\sum_g \|k_g\| = 2^D + 1$$

$\Rightarrow 2^D + 1$  inequivalent irreps of  $C_D$  for  $D \in 2\mathbb{Z}$

Claim: The # of inequivalent 1-D reps of any finite group  $G$  is equal to  $\|G\| / \|\text{Com}(G)\|$ .

The Commutator group is defined as follows:

$$\text{Com}(G) := \{aba^{-1}b^{-1} \mid a, b \in G\}$$

$\Rightarrow$  For  $D = 2\mathbb{Z}$ ,  $\text{Com}(C_D) = \{1, -1\} \cong \mathbb{Z}_2$

$$\begin{aligned} \text{(\# of 1-D irreps of } C_D) &= \frac{\|C_D\|}{\|\text{Com}(C_D)\|} = \frac{2^{D+1}}{2} \\ &= 2^D \end{aligned}$$

\* Now Since the total number of irreps is  $2^D + 1$  we conclude:

$\rightarrow$  There is only one irreducible rep of  $C_D$ , whose  $\dim(\rho) > 1$ .

• Suppose  $G$  has  $p$  inequivalent irreps of dimension  $n_p$ , then

$$\|G\| = \sum_p (n_p)^2, \quad p = 1, \dots, 2^D + 1$$

$$\|C_D\| = 2^{D+1} = \sum_{p=1}^{2^D} 1 + n^2$$

$$n = \dim(\rho) \neq 1$$

$$\dim C_D = 2^{D+1} = 2^D + n^2$$

∴

$$n = \dim(\rho) = 2^{D/2}$$

Thm: For  $D$  even, the group  $C_D$  has  $2^{D+1}$  irreps. Of these  $2^D$  are 1D and the remaining has dim  $2^{D/2}$

\* As such,

$$\rho: C_D \rightarrow \text{End}(V)$$

$$\dim(V) = 2^{D/2}$$

So we can expand  $\gamma_m$  as a  $2^{D/2} \times 2^{D/2}$  element of  $GL_{\mathbb{R}}(2^{D/2})$

∴ The above irreps are not irreps of  $\text{Cliff}(C_D)$  and only of the group  $C_D$  as  $C_D$  is the multiplicative group  $(C_D, \cdot)$ .  $\text{Cliff}(C_D)$  is defined over  $\mathbb{R}$  as well.

\* Notation: We use  $\gamma_m$  for both the elements of  $C_D$  as well as the  $2^{D/2}$  irrep  $\rho: G \rightarrow \text{End}(V)$

$$\therefore \rho(\gamma_m) = \tilde{\gamma}_m$$

(Don't use  $\sim$ )

\*  $\gamma_m^*$  satisfies  $\{\gamma_m^*, \gamma_n^*\} = 2\delta_{mn} \mathbb{I}$

Since

$$(\gamma_m \gamma_n)^* = \gamma_n^* \gamma_m^*$$

$$\therefore \{\gamma_m, \gamma_n\}^* = 2\eta_{mn} \mathbb{I} = \{\gamma_m^*, \gamma_n^*\} = 2\eta_{mn} \mathbb{I}$$

Now, since we have established  $\exists$  only one irrep of  $C_0$  w/  $\dim(\rho) > 1$  then

$$\gamma_m^* \sim \gamma_m$$

Two irreps are equivalent if  $\exists B \in GL(\mathbb{R}^1)$

s.t.  $\gamma_m^* = B \gamma_m B^{-1}$  (via Schur's lemma).

Take  $B \in SL(2^{D/2}, \mathbb{C})$

$$\begin{aligned} \gamma_m &= (B \gamma_m B^{-1})^* = B^* \gamma_m^* (B^{-1})^* \\ &= B^* B \gamma_m B^{-1} (B^{-1})^* \end{aligned}$$

$$\therefore B B^* \gamma_m = \gamma_m B B^*$$

Now, by Schur's lemma

$$B B^* = E \mathbb{I}$$

$$B^* = E B^{-1}$$

$$\leadsto (B B^*) (B B^{-1}) = E^* \mathbb{I}$$

$$B (B^* B) B^{-1} = B E B^{-1} = E \mathbb{I}$$

$$E = E^*$$

$$\therefore |B^*| = 1 \Rightarrow |E| = 1$$

$$E = \pm 1$$

$$B B^* = \pm \mathbb{I}$$

Now, consider the transpose  $\gamma^T$ .

$$\{\gamma_m, \gamma_n\}^T = 2\eta_{mn} \mathbb{I}$$

$$\{\gamma_m^T, \gamma_n^T\} = 2\eta_{mn} \mathbb{I}$$

$$(\gamma_m \gamma_n)^T + (\gamma_n \gamma_m)^T = 4\delta_{nm} \mathbb{I}$$

$$\gamma_n^T \gamma_m^T + \gamma_m^T \gamma_n^T = \dots$$

Thus, by Shur's lemma  $\exists C \in GL_{2D}(\mathbb{C})$   
 s.t.  $\boxed{\gamma_m^T = -C \gamma_m C^{-1}}$  | C: Charge Conjugation Matrix

that is to say,  $\gamma_m^T \sim (-\gamma_m)$

Assume  $M = \mathbb{M}^D$ ,  $\eta = \text{diag}(-1, +1, \dots, +1)$   
 $\Rightarrow$  Any finite-dim irrep of a finite group can be chosen to be unitary:

$$\boxed{\gamma_m \gamma_m^\dagger = \mathbb{I}}$$

Note  $\gamma_n \gamma_n = \mathcal{K}_{nn}$ .

$$\boxed{\gamma_0} \quad \left[ \begin{array}{l} \gamma_0^2 = -1 \\ \gamma_0 \gamma_0^\dagger = 1 \end{array} \right] \Rightarrow \boxed{\gamma_0^\dagger = -\gamma_0}$$

$$\boxed{\gamma_m} \quad \left[ \begin{array}{l} \gamma_m \gamma_m = 1 \\ \gamma_m \gamma_m^\dagger = 1 \end{array} \right] \Rightarrow \boxed{\gamma_m^\dagger = \gamma_m, m=1, \dots, D-1}$$

Combined this is also read as

$$\boxed{\gamma_m^\dagger = \gamma_0 \gamma_m \gamma_0}$$

Proof:

$$\left[ \begin{array}{l} \gamma_m \gamma_n + \gamma_n \gamma_m = 2\delta_{mn} \mathbb{I}, m \neq 0 \\ \gamma_0 \gamma_m + \gamma_m \gamma_0 = 0 \\ \Rightarrow \gamma_0 \gamma_m = -\gamma_m \gamma_0 \\ \Rightarrow \gamma_0 \gamma_m \gamma_0 = -\gamma_m \gamma_0^2 \\ \therefore \boxed{\gamma_m = \gamma_0 \gamma_m \gamma_0} \end{array} \right.$$

$$\gamma_m^\dagger = B \gamma_m B^{-1}$$

$$\gamma_m^T = (B^{-1})^T \gamma_m^T B^T = -(B^{-1})^T C \gamma_m C^{-1} B^T$$

$$= \gamma_0 \gamma_m \gamma_0$$

$$\Rightarrow \gamma_0 = C^{-1} B^T \quad | \quad -(B^{-1})^T C = \gamma_0$$

$$C \gamma_0^2 = B^T \gamma_0 \quad | \quad -B^T (B^{-1})^T C = B^T \gamma_0$$

$$\therefore C = -B^T \gamma_0 \quad | \quad \therefore C = -B^T \gamma_0$$

\* Place Further restrictions on B:

$$\gamma_m^T = (\gamma_m^*)^\dagger = (\gamma_m^\dagger)^*$$

$$\begin{aligned} \text{(LHS)} &= (B \gamma_m B^{-1})^\dagger \\ &= ((B^{-1})^T \gamma_m^T B^T)^* \\ &= (B^{-1})^\dagger \gamma_m^\dagger B^\dagger \\ &= (B^{-1})^\dagger \gamma_0 \gamma_m \gamma_0 B^\dagger \end{aligned}$$

$$\begin{aligned} \text{(RHS)} &= (\gamma_0 \gamma_m \gamma_0)^* \\ &= \gamma_0^* \gamma_m^* \gamma_0^* \\ &= (B \gamma_0 B^{-1}) (B \gamma_m B^{-1}) (B \gamma_0 B^{-1}) \\ &= B \gamma_0 \gamma_m \gamma_0 B^{-1} \end{aligned}$$

$$\gamma_0 \gamma_m \gamma_0 B^\dagger = B^\dagger B \gamma_0 \gamma_m \gamma_0 B^{-1}$$

$$- \gamma_m \gamma_0 B^\dagger B = \gamma_0 B^\dagger B \gamma_0 \gamma_m \gamma_0$$

$$\gamma_m (\gamma_0 B^\dagger B \gamma_0) = (\gamma_0 B^\dagger B \gamma_0) \gamma_m$$

$\therefore$  By Schur's lemma

$$\gamma_0 B^\dagger B \gamma_0 = \omega \mathbb{I}$$

$$\Rightarrow B^\dagger B = \mu \mathbb{I}$$

Since  $|B| = 1 \Rightarrow |\mu| = 1$

$$\langle N, N \rangle \geq 0$$

$$\therefore \langle B^\dagger B N, N \rangle = \langle B N, B N \rangle > 0$$

$$\therefore \bar{\mu} > 0, \mu > 0$$

Now,

$$\begin{aligned} \langle B^\dagger B N, N \rangle &= \bar{\mu} \langle N, N \rangle = \langle B N, B N \rangle = \langle N, B^\dagger B N \rangle \\ &= \mu \langle N, N \rangle \end{aligned}$$

$$\therefore \bar{\mu} = \mu$$

We conclude  $\mu$  is real and positive and so B is unitary:  $B^\dagger B = \mathbb{I}, \mu = 1$

$$B B^T = \epsilon \chi \Leftrightarrow B^T B^T = \epsilon \chi = \epsilon B^T B = B^T (\epsilon B)$$

$$\boxed{B^T = \epsilon B}, \quad \begin{aligned} C &= -B^T \gamma_0 \\ C^T &= -\gamma_0^T B \\ &= C \gamma_0 C^{-1} B \\ &= B^T \gamma_0 (B^T)^{-1} B \\ &= B^T \gamma_0 \epsilon \\ &= \epsilon B^T \gamma_0 \end{aligned}$$

$$\therefore \boxed{C^T = -\epsilon C}$$

\* Determining  $\epsilon$  in terms of spacetime dim  $D$ :  
Consider the set:

$\chi, \gamma_{m_1}, \gamma_{m_1 m_2}, \gamma_{m_1 m_2 m_3}, \dots, \gamma_{m_1 \dots m_D}$   
which contains

$$2^D = \sum_p \binom{D}{p} = (1+1)^D$$

elements, all linearly independent and span  $GL(2^{D/2}, \mathbb{C})$

$$\text{Claim: } \boxed{\gamma_{m_1 \dots m_p} = (-1)^{p(p-1)/2} \gamma_{m_p \dots m_1}}$$

Proof:

$$\begin{aligned} \gamma_{m_1 \dots m_p} &= (-1)^{p-1} \gamma_{m_p m_1 \dots m_{p-1}} \\ &= (-1)^{p-1} (-1)^{p-2} \gamma_{m_p m_{p-1} m_1 \dots m_{p-2}} \\ &= (-1)^{p-1} \gamma_{m_p m_{p-1} \dots m_1} \\ &= (-1)^{p(p-1) - \sum_{i=1}^{p-1} i} \gamma_{m_p m_{p-1} \dots m_1} \end{aligned}$$

where  $\sum_{i=1}^{p-1} i = \frac{n(n+1)}{2}$   
 $\Rightarrow p(p-1) - \frac{(p-1)p}{2} = \frac{p(p-1)}{2}$

$$\therefore \gamma_{m_1 \dots m_p} = (-1)^{p(p-1)/2} \gamma_{m_p m_{p-1} \dots m_1}$$

$$\begin{aligned} \gamma_{m_1 m_2}^T &= \frac{1}{2} (\gamma_{m_1} \gamma_{m_2} - \gamma_{m_2} \gamma_{m_1})^T \\ &= \frac{1}{2} (\gamma_{m_2}^T \gamma_{m_1}^T - \gamma_{m_1}^T \gamma_{m_2}^T) \\ &= \frac{1}{2} C (\gamma_{m_2} \gamma_{m_1} - \gamma_{m_1} \gamma_{m_2}) C^{-1} \\ &= -C \gamma_{m_1 m_2} C^{-1} \end{aligned}$$

$$\begin{aligned}
Y_{m_1, m_2, m_3}^T &= \frac{1}{6} (Y_{m_1} Y_{m_2} Y_{m_3} + \dots) \\
&= \frac{1}{6} (Y_{m_3}^T Y_{m_2}^T Y_{m_1}^T + \dots) \\
&= - \left( C \frac{1}{6} \right) (Y_{m_3} Y_{m_2} Y_{m_1} + \dots) C^{-1} \\
&= - C Y_{m_3, m_2, m_1} C^{-1} \\
&= (-1)^3 C [C^{-1}]^3 C Y_{m_1, m_2, m_3} C^{-1} \\
Y_{m_1, \dots, m_p}^T &= \frac{1}{p!} (Y_{m_1} \dots Y_{m_p} + \dots) \\
&= C (-1)^{\frac{p-1}{2}} \frac{1}{p!} (Y_{m_p} \dots Y_{m_1} + \dots) C^{-1} \\
&= (-1)^p C Y_{m_p, \dots, m_1} C^{-1} \\
&= (-1)^p (-1)^{p(p-1)/2} C Y_{m_1, \dots, m_p} C^{-1}
\end{aligned}$$

we conclude

$$C Y_{m_1, \dots, m_p} C^{-1} = (-1)^p (-1)^{p(p-1)/2} Y_{m_1, \dots, m_p}^T$$

Equivalently,

$$C Y_{m_1, \dots, m_p} = (-1)^p (-1)^{p(p-1)/2} Y_{m_1, \dots, m_p}^T C$$

$$\text{but } C^T = -E C \Leftrightarrow C = -E C^T$$

$$C Y_{m_1, \dots, m_p} = (-1)^p (-1)^{p(p-1)/2} E (C Y_{m_1, \dots, m_p})^T (-1)^{\frac{p-1}{2}}$$

Use  $(-1)^p = (-1)^{-p}$ , then

$$(-1)^{-(p-1)} (-1)^{p(p-1)/2}$$

$$C \rightarrow \frac{1}{2} (-2(p-1) + p(p-1)) = \frac{1}{2} (p-1)(p-2)$$

$$C Y_{m_1, \dots, m_p} = (-1)^{\frac{(p-1)(p-2)}{2}} E (C Y_{m_1, \dots, m_p})^T$$

likewise we have

$$Y_{m_1, \dots, m_p} C^{-1} = E (-1)^{(p-1)(p-2)/2} (Y_{m_1, \dots, m_p} C^{-1})^T$$

\* We can use this result to determine the # of anti-symmetric matrices in  $C_D$  when multiplied by  $C$ :

$$C : C_D \rightarrow ? \leftarrow (\text{may not be anti-sym})$$

Q: How many anti-symmetric matrices exist for  $C Y_{m_1, \dots, m_p}$ ?

→ My Q: why do we care?

$$A: \sum_{p=0}^D \frac{1}{2} (1 + E (-1)^{(p-1)(p-2)/2}) \binom{D}{p}$$

Weight Variables for  $E(-1)^{D+1} = 1$   
 i.e.  $(CY)$  is symmetric

Claim:  $(-1)^{(p-1)(p-2)/2} = -\frac{1}{2} [(1+i)i^p + (1-i)(-i)^p]$

~~Proof~~: Use induction

\* The number of  $2^{D/2} \times 2^{D/2}$  matrices is given by the number of independent elements

•  $n \times n$  - matrix which is anti-symm

$$(n^2 - n)/2 = n(n-1)/2$$

Now,  $n = 2^{D/2}$  so there are  $\frac{1}{2} 2^{D/2} (2^{D/2} - 1)$

$$\sum_{p=0}^D \frac{1}{2} \left( 1 + \frac{E}{2} [(1+i)i^p + (1-i)(-i)^p] \right) \binom{D}{p}$$

$$= \frac{1}{2} 2^D + \frac{E}{2} \left[ (1+i) \sum_{p=0}^D i^p \binom{D}{p} + (1-i) \sum_{p=0}^D (-i)^p \binom{D}{p} \right]$$

$$i^0 = 1$$

$$i^1 = i \quad \sum_{p=0}^D i^p \frac{D!}{(D-p)!p!}$$

$$i^2 = -1 \quad \dots$$

$$i^3 = -i$$

$$i^4 = 1$$

$$\frac{1}{2} 2^D + \frac{E}{4} ( ) = \frac{1}{2} (2^D - 2^{D/2})$$

$$\therefore \frac{E}{2} ( ) = -2^{D/2}$$

$$\text{or } E = -(2^{D/2}/2) ( )$$

$$\text{where } ( ) = \frac{1}{\sqrt{2}} [(1+i)^{D+1} + (1-i)^{D+1}]$$

$$\therefore E = -\sqrt{2} \cos \left\{ \frac{\pi}{4} (D+1) \right\}$$

$$\Rightarrow \cos \left[ \frac{\pi}{4} (D+1) \right] = \frac{1}{2} \left( e^{-i \frac{\pi}{4} (D+1)} + e^{i \frac{\pi}{4} (D+1)} \right)$$

$$= \frac{1}{2} \left( (e^{-i \pi/4})^{D+1} + (e^{i \pi/4})^{D+1} \right)$$

$$= \frac{1}{2} (\sqrt{2})^{-(D+1)} [(1+i)^{D+1} + (1-i)^{D+1}]$$

$$= \frac{2^{-D/2}}{2 \sqrt{2}} [(1+i)^{D+1} + (1-i)^{D+1}]$$

$$\text{Or, } E = \begin{cases} +1, & D = 2, 4 \pmod{8} \\ -1, & D = 0, 1 \pmod{8} \end{cases}$$

$$D \equiv 1, D \equiv 6, 8 \pmod{8}$$

Proof: 
$$e = 1 \quad - \frac{1}{\sqrt{2}} = \cos\left(\frac{\pi}{4}(D+1)\right)$$

$$\therefore \frac{\pi}{4}(D+1) = \frac{5\pi}{4} + 2\pi n$$

$$\Rightarrow D = 4 + 8n, \quad n \in \mathbb{N}$$

$$\text{or } \frac{\pi}{4}(D+1) = \frac{3\pi}{4} + 2\pi n$$

$$D = 2 + 8n$$

and likewise for

Since  $B^T = \epsilon B$ , for  $D \equiv 2, 4 \pmod{8}$

$$B^T = B \quad (\text{symmetric})$$

we already know  $B$  is unitary ( $B^\dagger B = \mathbb{I}$ )

$$\Rightarrow B = B_1 + iB_2, \quad B_1^T = B_1, \quad B_2^T = -B_2$$

$$B^\dagger B = (B_1 + iB_2)(B_1 - iB_2) = \mathbb{I}$$

$$= B_1^2 + B_2^2 + i(B_2 B_1 - B_1 B_2) = \mathbb{I}$$

$$B_1^2 + B_2^2 = \mathbb{I}$$

$$[B_1, B_2] = 0$$

\* Under a change of basis  $(\gamma^m)' = A \gamma^m A^{-1}$

$$(\gamma^{m*})' = A^* \gamma^{m*} A^{-1}$$

$$B' \gamma^{m'} B^{-1*} = A^* B \gamma^m B^{-1} A^{*-1}$$

$$= A^* B (A^{-1} \gamma^m A) B^{-1} A^{*-1}$$

$\therefore$

$$B' = A^* B A^{-1}$$

Can use  $A$  to diagonalize  $B$  s.t.

$$B = \mathbb{I} \quad D \equiv 2, 4 \pmod{8}$$

For  $D \equiv 2, 4 \pmod{8}$

(i) We can choose  $\gamma^m$  to be real ( $B = \mathbb{I}$ )

1 (i)  $\epsilon = 1$ ,  $C = \gamma_0$

\* Note  $C = -B^T \gamma_0$ ,  $B^T = \epsilon B = B$

however,  $\gamma^0 = \eta^{00} \gamma_0 = (-1) \gamma_0$ ,  
which is where the (-1) goes.

Manipulation of  $\gamma$ 's:  
 $\gamma_p \gamma_m$

- Can all be different:  $p \neq n \neq m$   $(\gamma_a)^2 = \pm 1$
- Two the same:  $\gamma_a \gamma_a$

$$\gamma_p \gamma_{m_1 \dots m_q} = \gamma_{p m_1 \dots m_q} + q \eta_{p m_1} \gamma_{m_2 \dots m_q}$$

$$[\frac{i}{2} \gamma_{mn}, \gamma_p] = \eta_{pn} \gamma_m - \eta_{pm} \gamma_n$$

$$\gamma_{a_1 \dots a_p} \gamma_{b_1 \dots b_q} = \sum_r (-1)^{r(r+1)/2} \frac{p! q!}{r! (p-r)! (q-r)!} \sum_{[a_1 \dots a_r] [b_1 \dots b_r]} \gamma_{a_1 \dots a_r b_1 \dots b_r} \gamma_{b_{r+1} \dots b_q}$$

Spinors in Even Dim:

$\text{Spin}(1, D-1)$ : Generated by  $\frac{i}{2} \gamma_{mn}$ , i.e.

$\hat{\mathcal{C}}_D$  is the Lie Alg. of  $\text{Spin}(1, D-1)$

$\leadsto$  If  $g \in \text{Spin}(1, D-1)$ , then

$$g = e^{\frac{i}{2} \omega^{mn} T_{mn}} = \mathbb{1} + \frac{i}{2} \omega^{mn} T_{mn} \quad \left( \begin{array}{c} T_{mn} = \frac{i}{2} \gamma_{mn} \\ \mathbb{1} \end{array} \right)$$

$$= \mathbb{1} + \frac{i}{4} \omega^{mn} \gamma_{mn}$$

Where  $\omega^{mn} = -\omega^{nm}$

Assume  $\text{Spin}(1, D-1)$  is connected to the identity  
i.e.  $\exists$  a continuous path from any  $g$  to  $\mathbb{1}$

$$\left. \frac{d}{dt} (e^{t \omega^{mn} \gamma_{mn}}) \right|_{t=0} = \omega^{mn} \gamma_{mn}$$

$\leadsto \det(A) = \exp(\text{Tr}(\ln(A)))$

$$A = \exp\left(\frac{i}{4} \omega^{mn} \gamma_{mn}\right)$$

$$\ln(A) = \frac{i}{4} \omega^{mn} \gamma_{mn}$$

$$\det(A) = \exp\left(\text{Tr}\left(\frac{i}{4} \omega^{mn} \gamma_{mn}\right)\right)$$

$$= \exp\left(\frac{i}{4} \omega^{mn} \text{Tr}(\gamma_{mn})\right)$$

$$\text{Tr}(\gamma_{mn}) = \frac{1}{2} (\text{Tr}(\gamma_m \gamma_n) - \text{Tr}(\gamma_n \gamma_m))$$

$$= \frac{1}{2} (\text{Tr}(\gamma_m \gamma_n) - \text{Tr}(\gamma_m \gamma_n))$$

$$= 0$$

$$\therefore \det(A) = 1$$

\* For an infinitesimal spin transformation an element of  $C_p$  of the form  $V = V^m \gamma_m$  are preserved in form under

$$V \rightarrow V' = A V A^{-1}$$

for any spin transformation  $A$ :

$$V' = \left(1 + \frac{i}{4} \omega^{mn} \gamma_{mn}\right) (V \gamma_p) \left(1 - \frac{i}{4} \omega^{ab} \gamma_{ab}\right)$$

$$= (V \gamma_p + \frac{i}{4} \omega^{mn} \gamma_{mn} V \gamma_p) \left(1 - \frac{i}{4} \omega^{ab} \gamma_{ab}\right)$$

$$= V \gamma_p + \frac{i}{4} \omega^{mn} \gamma_{mn} V \gamma_p - \frac{i}{4} V \gamma_p \omega^{ab} \gamma_{ab}$$

$$= V \gamma_p + \frac{i}{2} (V \gamma_p \omega^{mn}) \left(\frac{1}{2} \gamma_{mn} \gamma_p - \gamma_p \frac{1}{2} \gamma_{mn}\right)$$

Now, above (5.2.23) we have

$$\left[\frac{1}{2} \gamma_{mn}, \gamma_p\right] = \eta_{pn} \gamma_m - \eta_{pm} \gamma_n$$

$$V' = V \gamma_p + \frac{i}{2} (V \gamma_p \omega^{mn}) (\eta_{pn} \gamma_m - \eta_{pm} \gamma_n)$$

$$= V \gamma_p + \frac{i}{2} V_n \omega^{mn} \gamma_m - \frac{i}{2} V_m \omega^{mn} \gamma_n$$

$$= V + \frac{i}{2} V_m \omega^{nm} \gamma_n + \frac{i}{2} V_m \omega^{nm} \gamma_n$$

$$= V + V_m \omega^{nm} \gamma_n$$

$$= V^n \gamma_n + (\omega^{nm} V_m) \gamma_n$$

$$= (V^n + \omega^n_m V^m) \gamma_n$$

$$\therefore V'^n = V^n + \omega^n_m V^m \quad \left( \text{Infinitesimal Lorentz transformation} \right)$$

$$e^B e^{-B} = 1 + [B, C] + \frac{1}{2} [B, [B, C]] + \dots$$

from which we can conclude

$$V \rightarrow V' = \Lambda V \Lambda^{-1}$$

preserves the form of  $V$  for any spin transformation  $\Lambda$  and induces a finite Lorentz trans. on  $V^\mu$ .

•  $[\sum_i (\pi \gamma_i), V]$

~> Alternatively, we can view

$$\text{Spin}(1, D-1) = \left\{ \sum_{n_1, \dots, n_p} c_{n_1, \dots, n_p} \gamma^{n_1, \dots, n_p} = A \mid e^A V e^{-A} \text{ is preserved} \right\}$$

$V = V^\mu \gamma_\mu$

Thm:  $\text{Spin}(1, D-1) / \mathbb{Z}_2 \cong \text{ISO}^+(1, D-1)$

Proof:

$A \in \text{Spin}(1, D-1)$ , we can induce a linear map on  $\mathbb{R}^{(1, D-1)}$

•  $V'^\mu = V^\mu + \omega^\mu{}_\nu V^\nu$

for  $V = V^\mu \gamma_\mu$

$$\phi: \text{Spin}(1, D-1) \xrightarrow{\sim} \mathbb{R}^{(1, D-1)}$$

$$\phi(V) = \phi(V^\mu \gamma_\mu) = V^\mu \phi(\gamma_\mu)$$

$$\phi(\gamma_\mu) := \phi_\mu$$

s.t.  $\mathbb{R}^{(1, D-1)} = \text{span} \{ \phi_\mu \}_{\mu=1}^D$

$$\phi(\gamma_\mu) = e_\mu$$

$$\phi = \phi^\mu{}_\nu \Rightarrow \phi^\mu{}_\nu \gamma_\mu = e_\nu$$

~>

Now, we use this to take

$$\begin{aligned} \phi(e^A V) &= \phi \left[ \left( 1 + \frac{1}{4} \omega^{\mu\nu} \gamma_{\mu\nu} \right) V^\rho \gamma_\rho \right] \\ &= V^\rho e_\rho + \frac{1}{4} \omega^{\mu\nu} V^\rho \phi(\gamma_{\mu\nu} \gamma_\rho) \\ &= V^\rho e_\rho + \frac{1}{4} \omega^{\mu\nu} V^\rho \left( \phi(\gamma_{\mu\nu} \gamma_\rho) \right) \\ &\quad + \left( \eta_{\rho\mu} \gamma_\nu - \eta_{\rho\nu} \gamma_\mu \right) \end{aligned}$$

$$\begin{aligned}
 &= V^p e_p + \frac{1}{4} \omega^{mn} V^p [\phi(\gamma_{mnp})] \\
 &\quad + \phi_{pm} e_n - \phi_{pn} e_m \\
 &= V^p e_p + \phi_{mnp} + \frac{1}{4} \omega^{mn} V_m e_n - \frac{1}{4} \omega^{mn} V_n e_m \\
 &= V^p e_p + \phi_{mnp} + \frac{1}{2} \omega^{mn} V_m e_n
 \end{aligned}$$

...  
We write

$$V'^m = \Lambda^n_m V^n$$

• Note:

$$2^{D/2} V'^m V'_m = \text{Tr}(V'V') = \text{Tr}(VV) = 2^{D/2} V^m V_m$$

$$\text{Tr}(VV) = V^m V^n \text{Tr}(\gamma_m \gamma_n)$$

$$\text{Tr}(\gamma_m \gamma_n + \gamma_n \gamma_m) = 2 \eta_{mn} \text{Tr}(\mathbb{I})$$

$$\text{Tr}(\gamma_m \gamma_n) + \text{Tr}(\gamma_n \gamma_m) = 2 \eta_{mn} 2^{D/2}$$

$$\text{Tr}(\gamma_m \gamma_n) = 2^{D/2} \eta_{mn}$$

$$\therefore \text{Tr}(VV) = 2^{D/2} V^m V_m$$

$\leadsto$  Lorentz Group:  $\Lambda^n_m$  s.t.

$$\boxed{\Lambda \eta \Lambda^T = \eta}$$

The map

$$A \mapsto \Lambda_A$$

$$V_1 = A_1 V A_1^{-1}, \quad V_2 = A_2 V A_2^{-1}$$

then

$$V \xrightarrow{A_1} V_1 \xrightarrow{A_2} V_{12}$$

$$V \mapsto A_1 V A_1^{-1} \mapsto A_2 (A_1 V A_1^{-1}) A_2^{-1}$$

$$\begin{aligned}
 V_{12} &= (A_2 A_1) V (A_2 A_1)^{-1} \\
 &= A_2 V_1 A_2^{-1}
 \end{aligned}$$

$$\therefore \boxed{V_{12}^m = \Lambda_{A_2 n}^m \Lambda_{A_1 p}^n V^p}$$

$$A_1, A_2 \mapsto \Lambda_{A_1}, \Lambda_{A_2}$$

$\leadsto$  The above map is a homomorphism

Kernel of  $A$ :  $\{A' | A \rightarrow \mathbb{I}\}$

$$A \gamma_n = \gamma_n A$$

$$\bullet \text{ 1 commutator: } [A, \gamma_n] = A \gamma_n - \gamma_n A = 0$$

$\Lambda$  commutes with  $\mathbb{C}D$ ,  $A = \mathbb{C}A$ ,  $\mathbb{C} = -1$   
 $(\text{Kern} : \pm \mathbb{Z}) \cong \mathbb{Z}_2$

$\Rightarrow \det(\Lambda_A) = \pm 1$  (from  $\Lambda \Lambda^T = \mathbb{I}$ )

$A \mapsto \det(\Lambda_A)$

$\text{Spin}(1, D-1) \rightarrow \mathbb{R}$

• Since this map is continuous  $\exists$  a path from  $\mathbb{I}$  of  $\text{Spin}(1, D-1)$

$$(\Lambda_0)^2 = 1 + (\sum_i \Lambda_i^2) \geq 1$$

$$SL(2, \mathbb{C}) / \mathbb{Z}_2 \cong SO(1, D-1)$$

$$x'^n \gamma_n = A x^n \gamma_n A^{-1}$$

$\Rightarrow$  Use to induce Lorentz transf. on the coordinates of  $M^D$ .

$$\Lambda_a x^a \gamma_n = A x^n \gamma_n A^{-1}$$

\* A spinor  $\lambda$  transforms under  $\text{Spin}(1, D-1)$  as

$$\delta \lambda = \frac{1}{4} \omega^{mn} \gamma_{mn} \lambda$$

by definition,  $\omega^{mn} = -\omega^{nm}$  are the parameters of the infinitesimal spin transformation.

• Under a finite transformation

$$\lambda' = A \lambda$$

$\gamma_n$ :  $2^{D/2} \times 2^{D/2}$  matrices

$\Rightarrow \lambda$ : has  $2^{D/2}$  components

$\gamma^n \partial_n \lambda = 0$  is invariant under  $\delta$ , as well as finite  $\text{Spin}(1, D-1)$  transformations.

• An infinitesimal  $\text{Spin}(1, D-1)$  transformation induces a Lorentz transformation

$$x'^n = x^n + \omega^n_m x^m, \text{ on } M^D$$

$$\Rightarrow \partial'_n = \partial x^\ell / \partial x'^n \partial_\ell$$

$$\partial \lambda = 0$$

$$\frac{\partial}{\partial x'^n} = \frac{\partial \lambda}{\partial x'^n} \frac{\partial}{\partial x^\lambda}$$

$$x^n = e^{-\omega} x'^n = e^{\omega} x^n$$

$$x^n = e^{-\omega} x'^n = x'^n - \omega^n_m x'^m$$

• Proof:

$$x^n = x^n + \omega^n_\ell x^\ell - \omega^n_m (x^m + \omega^m_\ell x^\ell)$$

$$= x^n + \omega^n_\ell x^\ell - \omega^n_\ell x^\ell - \omega^n_m \omega^m_\ell x^\ell$$

$$\frac{\partial x^\ell}{\partial x'^n} = \delta_n^\ell - \omega_n^\ell \delta_n^m$$

$$\therefore \frac{\partial}{\partial x'^n} = (\partial_n^\ell - \omega_n^\ell) \partial_\ell$$

$$= \partial_n - \omega_n^\ell \partial_\ell$$

$$= \partial_n + \omega_n^m \partial_m$$

$$\Rightarrow \boxed{\partial'_n = \partial_n + \omega_n^m \partial_m}$$

With this we have:

$$\gamma^n \partial_n \lambda \rightarrow \gamma^n \partial'_n \lambda' = \gamma^n (\partial_n + \omega_n^\ell \partial_\ell) (\lambda + \frac{1}{4} \omega^{ma} \gamma_{ma} \lambda)$$

$$= \gamma^n \partial_n \lambda + \gamma^n \partial_n (\frac{1}{4} \omega^{ma} \gamma_{ma} \lambda) + \gamma^n \omega_n^\ell \partial_\ell \lambda$$

$$= \gamma^n \partial_n \lambda + \gamma^n (\frac{1}{4} \omega^{ma} \gamma_{ma}) \partial_n \lambda + \gamma^n \omega_n^\ell \partial_\ell \lambda$$

$$\gamma^n_{ma} = \gamma^n_{ma} + (\delta^n_m \gamma_a - \delta^n_a \gamma_m)$$

$$= \gamma^n \partial_n \lambda + \frac{1}{4} \gamma^n_{ma} \omega^{ma} \lambda + \frac{1}{4} \omega^{ma} (\delta^n_m \gamma_a - \delta^n_a \gamma_m) \lambda$$

$$+ \gamma^n \omega_n^m \partial_m \lambda$$

$$[\frac{1}{2} \gamma_{mn}, \gamma_p] = \frac{1}{2} \partial_{mn} \gamma_p - \gamma_p \frac{1}{2} \gamma_{mn} = \eta_{pn} \delta_m - \eta_{pm} \delta_n$$

$$\therefore \gamma_p \gamma_{mn} = \gamma_{ma} \gamma_p + 2 (\eta_{pm} \gamma_n - \eta_{pn} \gamma_m)$$

$$\frac{1}{4} \gamma^n \omega^{ma} \gamma_{ma} \partial_n \lambda$$

$$= \frac{1}{4} (\gamma_{ma} \gamma^n \omega^{ma} + 2 (\delta^n_a \gamma_m - \delta^n_m \gamma_a)) \partial_n \lambda$$

$$= \frac{1}{4} \omega^{ma} \gamma_{ma} (\gamma^n \partial_n \lambda) + \frac{1}{2} (\omega^{mn} \gamma_m - \omega^{nm} \gamma_m) \partial_n \lambda$$

$$= u \omega \quad \partial_m (\gamma^\mu \partial_\mu \lambda) = \omega^{\mu\nu} \partial_\mu \partial_\nu \lambda$$

$$\begin{aligned} \gamma^\mu \partial_\mu \lambda' &= \gamma^\mu \partial_\mu \lambda + \frac{1}{4} \omega^{\mu\nu} \gamma_{\mu\nu} (\gamma^\mu \partial_\mu \lambda) - \omega^{\mu\nu} \partial_\mu \partial_\nu \lambda + \dots \\ &= 0 \quad \checkmark \end{aligned}$$

Dirac Conjugate:  $\bar{\lambda}^D$ ,  $\bar{\lambda}^D \lambda \equiv \bar{\lambda}^{D\alpha} \lambda_\alpha$   
is also invariant, thus  $\bar{\lambda}^D = \lambda^\dagger \gamma_0$

$$\begin{aligned} \delta(\bar{\lambda}^{D\alpha} \lambda_\alpha) &= \delta \bar{\lambda}^{D\alpha} \lambda_\alpha + \bar{\lambda}^{D\alpha} \delta \lambda_\alpha = 0 \\ &= \delta \bar{\lambda}^{D\alpha} \lambda_\alpha + \bar{\lambda}^{D\alpha} \left( \frac{1}{4} \omega^{\mu\nu} \gamma_{\mu\nu} \lambda_\alpha \right) = 0 \\ \left( \delta \bar{\lambda}^{D\alpha} + \bar{\lambda}^{D\alpha} \left( \frac{1}{4} \omega^{\mu\nu} \gamma_{\mu\nu} \right) \right) \lambda_\alpha &= 0 \\ \lambda_\alpha &\neq 0, \text{ thus:} \end{aligned}$$

$$\boxed{(\delta \bar{\lambda}^D)^\alpha = (\bar{\lambda}^D)^\alpha \left( -\frac{1}{4} \omega^{\mu\nu} \gamma_{\mu\nu} \right)}$$

$$\begin{aligned} \text{Since } \gamma_{\mu\nu}^\dagger &= \gamma_0 \gamma_{\mu\nu} \gamma_0, \\ \delta(\lambda) &= \frac{1}{4} \omega^{\mu\nu} \gamma_{\mu\nu} \lambda \\ \delta \lambda^\dagger &= \frac{1}{4} \omega^{\mu\nu} \lambda^\dagger \gamma_{\mu\nu}^\dagger \\ &= \lambda^\dagger \frac{1}{4} \omega^{\mu\nu} \gamma_0 \gamma_{\mu\nu} \gamma_0 \\ \Rightarrow \delta \lambda^\dagger \gamma_0 &= (\lambda^\dagger \gamma_0) \left( \frac{1}{4} \omega^{\mu\nu} \gamma_{\mu\nu} \right) (-1) \\ &= (\lambda^\dagger \gamma_0) \left( -\frac{1}{4} \omega^{\mu\nu} \gamma_{\mu\nu} \right) \end{aligned}$$

$\therefore$  The Dirac Conjugate is defined as

$$\boxed{\bar{\lambda}^D = \lambda^\dagger \gamma_0}$$

Majorana Conjugate:  $\bar{\lambda}^M$

$$\boxed{\bar{\lambda}^M \equiv \lambda^T C}$$

$$\gamma_{\mu\nu}^T = -C \gamma_{\mu\nu} C^{-1}, \text{ and so}$$

$$\delta \lambda^T = \lambda^T \frac{1}{4} \omega^{\mu\nu} \gamma_{\mu\nu}^T$$

$$\Rightarrow \delta(\lambda^T C) = \lambda^T \left( -\frac{1}{4} \omega^{mn} C \gamma_{mn} C^{-1} \right) C$$

$$= (\lambda^T C) \left( -\frac{1}{4} \omega^{mn} \gamma_{mn} \right)$$

$\Rightarrow$  Transforms as the Dirac conj. under  $\text{Spin}(1, D-1)$

Majorana Spinor:  $\lambda$  is Majorana if  $\bar{\lambda}^D = \lambda^*$

$$\lambda^T \gamma^0 = \lambda^T C$$

$$\gamma^0{}^T \lambda^* = C^T \lambda$$

$$\lambda^* = -(\gamma^0)^T C^T \lambda$$

$$= B \lambda$$

$$\gamma^0{}^T = -C \gamma^0 C^{-1}$$

$$C^T = -C$$

$$\gamma^0{}^T \gamma^0{}^T = -1$$

$$C = B^T \gamma^0$$

$$\boxed{\lambda^* = B \lambda}$$

$$\Rightarrow \lambda = B^* \lambda^* = B^* B \lambda = \epsilon \lambda, \quad B^* B = \epsilon \mathbb{1}$$

Majorana Spinors only exist if  $\epsilon = +1$

$$D \equiv 2, 4 \pmod{8}$$

$$D = 2, 4, 10, 12, \dots$$

\* In even dmspacetime we can construct

$$\gamma^{D+1} \equiv \gamma_0 \gamma_1 \dots \gamma_{D-1} = \gamma_{01 \dots D-1}$$

$$\gamma^{D+1} \gamma_m = -\gamma_m \gamma^{D+1}$$

$$\Rightarrow \gamma^{D+1} \gamma_{mn} = \gamma^{D+1} \gamma_{mn} \text{ and so}$$

$$\gamma^{D+1} \left( \frac{1}{4} \gamma_{mn} \omega^{mn} \right) = \gamma^{D+1} \left( \frac{1}{4} \gamma_{mn} \omega^{mn} \right)$$

•  $\gamma^{D+1} \chi$  transforms as a spinor if  $\chi$  does  
i.e. if

$$\delta \chi = \frac{1}{4} \omega^{mn} \gamma_{mn} \chi, \text{ then}$$

$$\gamma^{D+1} \delta \chi = \frac{1}{4} \omega^{mn} \gamma^{D+1} \gamma_{mn} \chi = \frac{1}{4} \omega^{mn} \gamma_{mn} (\gamma^{D+1} \chi)$$

$$= \delta(\gamma^{D+1} \chi)$$

• Also,  $(\gamma^{D+1})^2 = (-1)^{(D+1)/2} \pm \mathbb{I}$

$$(\gamma^{D+1})^2 = \begin{cases} \mathbb{I} & D = 2 \pmod{4} \\ -\mathbb{I} & D = 4 \pmod{4} \end{cases}$$

Weyl Spinors:

$$\boxed{\begin{aligned} \gamma^{D+1} \chi &= \pm \chi & D = 2 \pmod{4} \\ i \gamma^{D+1} \chi &= \pm \chi & D = 4 \pmod{4} \end{aligned}}$$

$$\begin{aligned} \bar{\chi} \gamma_{m_1 \dots m_p} \lambda &= \lambda^T \alpha (C \gamma_{m_1 \dots m_p})^{\alpha\beta} \lambda_\beta \\ &= -(-1)^{(D-1)(p-2)/2} \frac{1}{\lambda} \gamma_{m_1 \dots m_p} \chi \end{aligned}$$

$$\bar{\chi} \gamma_n \lambda = -\bar{\lambda} \gamma_n \chi, \quad \bar{\lambda} \gamma_n \lambda = 0$$

↳ Proof:

$$\begin{aligned} \bar{\chi} \gamma_n \lambda &= \chi^T C \gamma_n \lambda \\ &= \chi^T C (-C^{-1} \gamma_n^T C \lambda) \\ &= -\chi^T \gamma_n^T C \lambda \\ &= \chi^T \gamma_n^T C^T (\lambda^T)^T \\ &= -(\lambda^T C \gamma_n \chi)^T \\ &= -\bar{\lambda} \gamma_n \chi \end{aligned}$$

From exchange of  $\chi \lambda \rightarrow \lambda \chi$

$$P_{\pm} = \frac{1}{2} (1 \pm a \gamma^{D+1}) \quad \Bigg| \quad a = \begin{cases} 1 & D = 2 \pmod{4} \\ -1 & D = 4 \pmod{4} \end{cases}$$

Central Charges:  $D = 2\mathbb{Z}$

$$\boxed{\{Q_\alpha, Q_\beta\} = (\gamma_m C^{-1})_{\alpha\beta} P^m + \sum_{P, P \neq 1} (\gamma_{m_1 \dots m_p} C^{-1})_{\alpha\beta} Z^{m_1 \dots m_p}}$$

$Z^{m_1 \dots m_p}$ : Central Charges

$P^m$ : translations Generator

- Note that the central terms can only exist if  $(\gamma_{m_1 \dots m_p} C^{-1})_{\alpha\beta}$  is symmetric in  $(\alpha\beta)$ .

Recall:  $C \gamma_{m_1 \dots m_p} C^{-1} = (-1)^P (-1)^{P(P-1)/2} \gamma_{m_1 \dots m_p}^T$

$C^T = -\epsilon C$  and so

$$\begin{aligned} \gamma_{m_1 \dots m_p} C^{-1} &= (-1)^P (-1)^{P(P-1)/2} C^{-1} \gamma_{m_1 \dots m_p}^T \\ &= (-1)^P (-1)^{P(P-1)/2} (\gamma_{m_1 \dots m_p} (C^{-1})^T)^T \\ &= (-1)^P (-1)^{P(P-1)/2} \epsilon (\gamma_{m_1 \dots m_p} C^{-1})^T \end{aligned}$$

$$\frac{2(P+1)}{2} + \frac{P(P-1)}{2} \Rightarrow P^2 - 3P + 2$$

$$\gamma_{m_1 \dots m_p} C^{-1} = \epsilon (-1)^{(P-1)(P-2)/2} (\gamma_{m_1 \dots m_p} C^{-1})^T$$

Now, the matrix is symmetric if:

$$1 = \epsilon (-1)^{(P-1)(P-2)/2}$$

$D \equiv 2, 4 \pmod{8}$ :  $\epsilon = 1$

$\Rightarrow P \equiv 1, 2 \pmod{4}$

$D \equiv 6, 8 \pmod{8}$ :  $\epsilon = -1$

$\Rightarrow P \equiv 3, 4 \pmod{4}$

Take  $Q_\alpha, Q_\beta$  to be Weyl

$$\{Q_\alpha, Q_\beta\} = (P \pm \gamma_m C^{-1})_{\alpha\beta} P^m + \sum_P (P \pm \gamma_{m_1 \dots m_p} C^{-1})_{\alpha\beta} Z^{m_1 \dots m_p}$$

Claim:  $(\gamma^{D-1} \gamma_{m_1 \dots m_p} C^{-1})_{\alpha\beta} = \epsilon (\gamma_{m_1 \dots m_{D-p}} C^{-1})_{\alpha\beta}$

$$\begin{aligned} (\gamma_{0 \dots p-1} (\gamma_{m_1 \dots m_p} C^{-1}))^T &= (\gamma_{m_1 \dots m_p} C^{-1})^T \gamma_{0 \dots p-1}^T \\ &= ( )^T C \gamma_{m_1 \dots m_p} C^{-1} (-1)^{D-1} C^{-1} \\ &= \epsilon (-1)^{(D-1)(D-2)/2} (-1)^{D-1} \gamma_{m_1 \dots m_p} C^{-1} C \gamma_{D-1 \dots 0} C^{-1} \end{aligned}$$

$$= \alpha \gamma_{m_1 \dots m_p} \gamma_{D-1 \dots 0} c^{-1}$$

$$\hookrightarrow \gamma_{m_1 \dots m_p} \gamma_{0 \dots D-1} = - \gamma_{m_1 \dots m_p} \gamma_{0 \dots D-1}$$

$$= - \sum_r (-1)^{r(r+1)/2} (-1)^{pr} \frac{p!(D-1)!}{r!(p-r)!(D-1-r)!} \Delta$$

$$\sum_{[m_1 \dots m_r] [m_{r+1} \dots m_p]} \gamma_{0 \dots D-1}$$

... Some how only the  $p$ th term survives...

$\rightarrow$  I believe the reasoning is as follows:

For  $r=0$ :

$$\gamma_{0 \dots D-1}$$

however  $p \leq D-1$  and so there will be duplicates i.e.

$$\gamma_{\dots i \dots i \dots} = 0$$

This goes on until  $r=p$  where all  $m_p$  are removed from the  $\gamma$ :

$$r=p:$$

$$= (-1)^{p(p+1)/2} \frac{p!(D-1)!}{p!(0)!(D-1-p)!} \sum_{[m_1 \dots m_p]} \gamma_{p \dots D-1}$$

$$= (-1)^{p(p+1)/2} \left[ \frac{(D-1)!}{(D-1-p)!} \right]$$

$$= (-1)^{p(p+1)/2} \epsilon_{0 \dots p} \epsilon_{m_1 \dots m_p} \frac{1}{p!} \gamma_{p+1 \dots D-1}$$

$$= (-1)^{p(p+1)/2} \epsilon_{m_1 \dots m_p} \gamma_{p+1 \dots D-1}$$

$\Rightarrow$  Relabel  $p+1 \rightarrow 1$  : i.e. subtract  $p$  from all terms:  $p \rightarrow p-1$

$$= (\epsilon \gamma_{m_1 \dots m_{p-p-1}} c^{-1})$$

$\sim$  NOT at all right but "close" I suppose.

||



Weyl Central Charges exist at rank  $p$

$$p \equiv 1, 2 \pmod{4}$$

$$D - p \equiv 1, 2 \pmod{4}$$

if

$$c \neq 1$$

Majorana-Weyl:  $D \equiv 2 \pmod{8}$

• Adopt the above conditions:

$$D = 2 + 8n, \quad n \in \mathbb{Z}$$

$$2 + 8n - p = 1 + 4m$$

$$2n - m = w$$

$$1 + (8n - 4m) = p$$

$$1 + 4(2n - m) = p$$

$$1 + 4w = p$$

← Okay

Now, assume  $D - p = 2 + 4m$

$$2 + 8n - p = 2 + 4m$$

$$p = 4(2n - m)$$

which is not allowed. Thus Maj.-weyl only exist in

$$D \equiv 2 \pmod{8}$$

$$p \equiv 1 \pmod{4}$$

ex:  $N=1, D=10$  susy

Type I supergravity in  $D=10$

$$\{Q_\alpha, Q_\beta\} = (P_\pm \gamma_m C^{-1})_{\alpha\beta} P^m + (P_\pm \gamma_{m_1 \dots m_5} C^{-1})_{\alpha\beta} Z^{m_1 \dots m_5} + (P_\pm \gamma_{m_1 \dots m_9} C^{-1})_{\alpha\beta} Z^{m_1 \dots m_9}$$

since no  $\gamma$ 's exist with  $p > 10$ .

$$\epsilon^{m_1 \dots m_9} \gamma_{m_1 \dots m_9} \propto \gamma^n \gamma_{11}$$

$$\{Q_\alpha, Q_\beta\} = (P_\pm \gamma_m C^{-1})_{\alpha\beta} P^m + (P_\pm \gamma_{m_1 \dots m_5} C^{-1})_{\alpha\beta} Z^{m_1 \dots m_5}$$

$D=11$  Majorana Supercharges

$$p \equiv 1, 2 \pmod{4}$$

$$\{Q_\alpha, Q_\beta\} = (\gamma^m C^{-1})_{\alpha\beta} P_m + (\Gamma^{mn} C^{-1})_{\alpha\beta} Z_{mn} + (\Gamma^{mnpqr} C^{-1})_{\alpha\beta} Z_{mnpqr}$$

$$\begin{aligned} \gamma^{m_1 \dots m_p} &= (-1)^{p(p-1)/2} (-1)^{q(q+1)/2} \epsilon^{m_1 \dots m_p n_1 \dots n_q} \gamma_{n_1 \dots n_q} \\ &= \frac{(-1)^{p(p+3)/2}}{(11-p)!} \epsilon^{m_1 \dots m_p n_1 \dots n_q} \gamma_{n_1 \dots n_q} \end{aligned}$$

$$p+q = 11, \quad \gamma_{10} = \gamma_0 \gamma_1 \dots \gamma_9$$

2D SUGRA:  $M_2, M_5$  branes

Type IIA: One Majorana Supersymmetry

$$\begin{aligned} \{Q_\alpha, Q_\beta\} &= (\gamma_m C^{-1})_{\alpha\beta} P^m + (\gamma_{m_1 m_2} C^{-1})_{\alpha\beta} Z^{m_1 m_2} + \Gamma^{(5)} \\ &\quad + \Gamma^{(6)} + \Gamma^{(7)} + \Gamma^{(10)} \end{aligned}$$